

Safe Route Mapping of Roadways Using Multiple Sourced Data

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Abstract—The use of systematic techniques with historical crash data and qualitative measures has long been a common practice to identify the problematic road features and develop countermeasures to mitigate the crash risk in crash-prone locations. This paper proposes a novel approach, *Safe Route Mapping (SRM)* model that integrates crash-based estimates with conflict risks computed from driver-based data to score the safety of roadways. An advanced Safety Performance Function (SPF) estimates the number of crashes, and a driver-based model computes dynamic conflict risk measures from driver and traffic data. In real-life implementations of the proposed methodology, the driver-based data and traffic data can be collected from vehicles or infrastructure-based data sources, including smartphones. We demonstrate the methodology using real historical crash data and simulated driver-based data obtained from VISSIM and SSAM. We show safety risk heat maps for the example roadway and illustrate how these maps change with driver types and traffic volumes. The proposed methodology fills the existing gaps in the use of near real-time dynamic data to designate safe corridors, dispatch law enforcement, and plan safety projects. Drivers can also use the road heat maps for situational awareness and trip planning.

Index Terms—Traffic safety, traffic conflict analysis, crash prediction, real-time risk scoring, safety performance function, dynamic risk heat mapping.

I. INTRODUCTION

HOT spot identification of roadways using systematic and systemic techniques with historical crash data and qualitative measures [1] has been commonly practiced by traffic authorities and law enforcement agencies for many years. The risk measures computed using these methods do not correctly reflect the fast-changing conditions in roadways, traffic, and weather, and other externalities. Moreover, these static techniques hardly account (aggregation at best) for

direct human factors and errors. Many reports consider human errors as the most significant contributing factors to traffic accidents [2]–[7].

This paper intends to fill this important gap by developing a data-driven methodology, hereafter, to be referred to as *Safe Route Mapping (SRM)*. This methodology combines *over-a-trip* safety data and historical crashes to develop *dynamic risk heat maps* of roadways. Onboard vehicle devices, smartphones, and infrastructure based IoT provide ample opportunities to collect *over-a-trip* safety data in real-time. These heat maps can be used by authorities to designate safe corridors, dispatch law enforcement, and strategize safety projects using hard near real-time dynamic data. At the same time, individual drivers can also benefit from having access to heat maps data for enhanced situational awareness. In this paper, we will focus on the *SRM* methodology and leave the discussion on the application topics, including the recent attempts by startups and technology companies to a separate article [8].

In principle, the *SRM* uses multiple data sources, e.g., historical crashes and roadway characteristics, driver-based data from smartphones or onboard devices, any existing systemic data on traffic safety, and dynamic roadway characteristics from IoT and social media. Here, we will only focus on a subset that includes historical crashes and simulated driver-based data (due to lack of actual data). For illustrative purposes, we will use a real roadway segment where actual crash data is available for a number of years [9]. Simulated conflicts will be calculated from the Surrogate Safety Assessment Model (SSAM) that operates on the traffic data from VISSIM simulation. The contributions of this paper are several folds: (i) An advanced Safety Performance Function (SPF) [10] considering multiple road features to estimate the number of crashes using real crash data. (ii) A methodology to create risk profiles of individual drivers with location and time attributes. A risk profile is a collection of risk measures that are sampled from a driver-based Machine Learning model. The model considers key driving safety behavioral attributes and road characteristics. (iii) Aggregation of risk profiles into statistical risk measures for a roadway segment and a given period of time. These statistical measures change with driver mix and traffic flow. (iv) Risk heat mapping of roadway segments or facilities using a hybrid measure composed of SPF estimates and aggregated statistical measures from drivers. We present an approach that translates the number of crashes and the

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probability of conflicts into fuzzy measures and uses fuzzy logic to determine a common denominator.

II. LITERATURE REVIEW

The American Association of State Highway Transportation Officials' (AASHTO) Highway Safety Manual (HSM) guidelines on predictive safety measures for hot-spot ranking and the follow-up diagnosis and countermeasures have been widely used across the United States by many states and municipalities [11]. A number of safety predictive models using crash data have been proposed and widely used in the literature and practice. SPF is a crash prediction model that provides an estimation of the expected average crash frequency of a network, facility, or individual site [11]. A crash modification factor (CMF), which could vary by site specifics and collision, is also applied to estimate the change in the expected number of crashes at the site when a specific countermeasure is implemented [12]. Crash frequency belongs to a general family of stochastic models that are often referred to as counting processes. The Poisson regression model is usually suggested to explain the relationship between crash-frequency with roadway and traffic characteristics [13], [14]. Researchers also use the Poisson-Gamma or Poisson-lognormal model/Negative Binomial (NB) model in which the Poisson parameter follows a gamma or lognormal distribution [15], [16]. Empirical Bayes (EB) method has been suggested to improve the quality of results [17]–[19]. The EB method uses a weight factor, which is a function of the SPF over-dispersion parameter, to combine the two estimates into a weighted average [20]. In cases when researchers may not have data to build their own SPF models, paper [21] explores methods to enhance the transferability of the developed SPF model.

The current practices of SPF-based predictions suffer from data integrity, lack of sufficient data, and model accuracy due to aggregation and calibration factors. To fill these gaps, risk-based approaches that take advantage of hard data, collected from vehicle and roadway sensors, slowly but surely reform the traffic safety literature. The SHARP-II Naturalistic Driving Behavior (NDB) and the subsequent safety studies [22]–[24] strongly support this point. With crashes being rare events, traffic conflicts are used as rough proxy or surrogates for traffic accidents in the Highway Safety Improvement Program [25]. In real-life traffic safety studies, Near-miss accidents often play this role. A near-miss accident can be seen as a vehicle conflict requiring immediate maneuvers to avoid a crash [26]. Near-misses were identified through a two-step data processing. Firstly, events were detected using predefined threshold values. If at least one of the threshold values, namely, Lateral Acceleration, Longitudinal Acceleration, Forward/Rear Time-To-Collision (TTC), or Yaw Rate, had been violated, or the driver had activated the Button, a near-miss accident was recorded [26]. Reference also [26] presents detailed information about threshold values and definitions. Several works exist in the literature use driver-based data to analyze collisions and crash surrogates, assess risks, and construct collision avoidance advisory systems [27]–[30]. Arbabzadeh and Jafari [31] developed an enhanced Advanced

Driver Assistance System using real-time safety risk measures predictions based on driver-based data. Using fuzzy logic, the relationships between risk sources and the consequences were identified and quantified in [32]. A systemic analysis in [33] uses a risk-based methodology to identify the relationship between roadway geometric features and crash types. The approach is qualitative and uses crash history on an aggregate basis.

The driver-based and vehicle-based data approaches have been suffering from low market penetration, which gives a non-reliable or low-density level of data. On the other hand, infrastructure sensing requires significant instrumentation via the conventional approach and becomes too costly for potential deployment [34], [35]. With the anticipated increase in the number of connected vehicles, smartphone applications [36]–[38], and onboard devices, driver-based or vehicle-based data will be of less challenge. However, there will still be data confidentiality and security issues to be addressed. To make up for the lack of data, micro traffic simulation software such as VISSIM and SUMO along with Surrogate Safety Assessment Model (SSAM) [39], has been widely used for traffic safety assessment due to its capability to model driver behaviors, vehicle movement, and interactions with infrastructures and other vehicles [40]. The simulation-based approach provides a reliable and insightful way to diagnose traffic safety problems proactively and to evaluate appropriate initiatives without getting into actual crashes [40].

III. DESCRIPTION OF SRM METHODOLOGY

The objective of this paper is to develop a methodology that creates dynamic risk heat mapping of road facilities (e.g., intersection, segment, ramps) considering multiple-sourced data such as historical crashes, roadway characteristics, vehicle-based data, and driver-based data. Actual applications using the *SRM* methodology include, but not limited to, Real-time Alert for drivers in high-risk situations due to dangerous behaviors such as drowsiness, lane departure, and speeding; Post-trip analysis for driving skill evaluation; and Safe Routes option in navigation using risk heat maps. Table I summarizes suggested data types for traffic safety studies and brings foundations for implementing the proposed *SRM* methodology [41].

Asides from basic data types such as speed, yaw rate, and acceleration, more advanced data types such as drowsiness, distraction, tailgating, and careless lane-change can now be detected by smartphone apps [37], [38]. We note that driver-based data changes fast in near real-time, but crash based data changes very slowly. In the absence of actual driver-based data, we will use VISSIM for traffic simulation and SSAM for traffic conflict analysis. SSAM computes surrogate measures of safety for each conflict that is identified in the trajectory data from VISSIM (see Appendix Table VII, VIII, and Fig. 16 for conflict angles and threshold values are taken from [42]. Fig. 1 shows the framework of the proposed *SRM* methodology.

The SPF model estimates crash counts for roadway segments or intersections using historical crash data.

TABLE I
SUGGESTED DATA FOR A TRAFFIC SAFETY STUDY

Internal		External	
Driver		Road	
Static	Demographics (age, gender, years of driving, etc.)	Static	Infrastructure, Road geometry, Historical crash
Semi-dynamic	Driving history (accident history, etc.)	Dynamic	Traffic flow, Traffic signal
Dynamic	Driver behavior (drowsy, distracted, texting, etc.)	Weather	
Vehicle		Dynamic	Precipitation, Light, Visibility
Static	Vehicle characteristics (model, make, etc.)	Time	
Dynamic	Speed, Acceleration	Dynamic	Day/night, Traffic peak hour

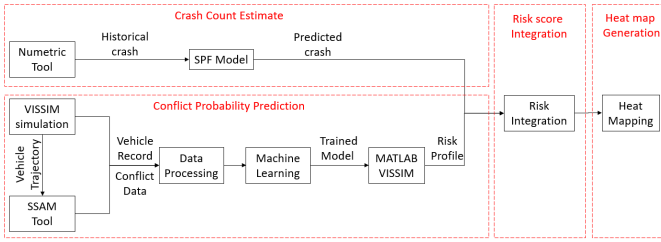


Fig. 1. The structure of the SRM methodology.

The Driver-based model predicts risk probabilities of individual drivers by considering both endogenous and exogenous risk factors. An Integration model is employed to fuse data from both models to obtain a single risk measure. In a real-life application of the SRM, there will be no need for simulation models.

A. Crash Prediction Model

In this section, an NB based SPF model is introduced for crash count prediction using our real crash data, and the EB method is used to improve the quality of results. With crashes being count data, they are often handled by Poisson regression for prediction [20]. Suppose $\{N_i(t), t \geq 0\}$ is the number of crashes in year t at site i . Many studies assume $N_i(t)$ a Poisson process [20] with parameter λ_i :

$$f(N_i(t), \lambda_i) = e^{-\lambda_i t} \frac{(\lambda_i t)^{N_i(t)}}{N_i(t)!} \quad (1)$$

The mean and variance of such a process are the same and equal to $\lambda_i t$. Since we would like to count crashes yearly, t is fixed to 1, and the mean and variance is reduced to λ_i . [43] suggests a Poisson regression model with a crash exposure variable for crash count prediction, which can be expressed as:

$$E(N_i(t)) = \lambda_i = VMT_i^\gamma e^{\sum_{j=0}^p \beta_j x_j} \quad (2)$$

Here, x_j 's are roadway features and traffic conditions for site i and β_j 's are coefficients. VMT is vehicle-miles traveled along the site i , and γ poses a potential nonlinear relationship between the crash count and VMT [43]. The exposure variable modifies the crash count to be weighted on the length of

the site. Under a special condition when all sites are uniformly separated, no modification is needed, and (2) can be reduced to:

$$E(N_i(t)) = e^{\sum_{j=0}^p \beta_j x_j} \quad (3)$$

The Poisson model assumes the mean equals the variance. Nevertheless, the variation of count data is often higher than the mean (i.e., over-dispersion) [20]. The over-dispersed data violates the Poisson assumption and thus make it not applicable to use Poisson regression for crash data. To overcome the over-dispersion, we assume λ_i is not fixed and follows a Gamma distribution:

$$\begin{aligned} f(N_i(t)|\lambda_i, v, \delta) &= \int_0^\infty e^{-\lambda_i} \frac{(\lambda_i)^{N_i}}{N_i!} G(\lambda_i|v, \delta) d\lambda_i \\ &= \int_0^\infty e^{-\lambda_i} \frac{(\lambda_i)^{N_i}}{N_i!} \frac{\delta^v}{\Gamma(v)} \lambda_i^{v-1} e^{-\delta \lambda_i} d\lambda_i \end{aligned} \quad (4)$$

where $G(\lambda_i|v, \delta)$ is Gamma distribution with parameter v and δ . Using the properties of Gamma distribution, (4) can be rewritten as:

$$\begin{aligned} f(N_i(t)|\lambda_i, v, \delta) &= \frac{\delta^v \Gamma(v + N_i) (1 + \delta)^{-(N_i+v)}}{\Gamma(v) \Gamma(N_i + 1)} \\ &\times \int_0^\infty \frac{e^{-\lambda_i(1+\delta)} (\lambda_i(1+\delta))^{N_i+v-1} (1+\delta)}{\Gamma(v + N_i)} d\lambda_i \end{aligned} \quad (5)$$

The outside part resembles a gamma distribution, and the integral part equals to 1. Therefore, (5) can be rewritten as:

$$f(N_i|v, \delta) = \frac{\Gamma(v + N_i)}{\Gamma(v) \Gamma(N_i + 1)} \left(\frac{\delta}{1 + \delta} \right)^v \left(\frac{1}{1 + \delta} \right)^{N_i} \quad (6)$$

This equation is similar to Type 2 Negative Binomial (NB2) distribution with parameters $r = v$ and $p = (1 + \delta)^{-1}$. The mean of NB2 can be expressed as $E(N_i) = \mu_i = \frac{v}{\delta}$ and the variance of NB2 can be expressed as $Var(N_i) = \mu_i + \frac{\mu_i^2}{v}$. It indicates that the variance is higher than the mean of crash counts. Therefore, NB2 can handle over-dispersion. Remember that μ_i^2 is the variance of the Poisson random variable. Therefore, $\frac{1}{v}$ (sometimes shows as a) is the magnitude of over-dispersion from a Poisson random variable. Using the variance of NB2, (6) can be tailored as:

$$\begin{aligned} f(N_i|a, \delta) &= \frac{\Gamma(\frac{1}{a} + N_i)}{\Gamma(\frac{1}{a}) \Gamma(N_i + 1)} \left(\frac{1}{1 + a\mu_i} \right)^{\frac{1}{a}} \\ &\times \left(1 - \frac{1}{1 + a\mu_i} \right)^{N_i} \end{aligned} \quad (7)$$

Since we will study the sites with uniform length, similar to (3), we can model the expected number of crashes as:

$$E(N_i) = \mu_i = e^{\sum_{j=0}^p \beta_j x_j} \quad (8)$$

β_j 's are estimated using the Maximum Likelihood Estimator (MLE) [44]. Using results from [45] on canonical link negative function and a number of transformations, we find a functional

form in terms of input variables (i.e., roadway characteristics, data from crash reports):

$$l(N_i|x_i, \beta, \alpha) = \sum_{i=1}^n \left(N_i \ln \frac{\alpha \exp(x'_i \beta)}{1 + \alpha \exp(x'_i \beta)} - \frac{\ln(1 + \alpha \exp(x'_i \beta))}{\alpha} + C_i \right) \quad (9)$$

where $\alpha = \frac{1}{v}$ and C_i is $\ln \Gamma(N_i + \frac{1}{\alpha}) - \ln \Gamma(N_i + 1) - n \Gamma(\frac{1}{\alpha})$. α and β can be obtained numerically [45]. Given estimates of β 's we calculate probabilities of crashes and necessary statistics on the number of crashes.

An effective way to improve crash predictive methods is to use past crash information, i.e., the EB method, to create a weighted average for improving crash prediction. From (5), the posterior distribution of λ_i follows a gamma distribution with parameters $v + N_i$ and $1 + \delta$. Thus, given historical data on N_i , we compute the conditional expected value of λ_i as below:

$$E(\lambda_i|N_i) = \frac{v + N_i}{1 + \delta}. \quad (10)$$

Recall $\alpha = \frac{1}{v}$ and $\delta = \frac{1}{\alpha \mu_i}$, (10) can be further expressed as:

$$E(\lambda_i|N_i) = \left(\frac{1}{1 + \alpha \mu_i} \right) \mu_i + \left(1 - \frac{1}{1 + \alpha \mu_i} \right) N_i \quad (11)$$

It shows that the expected number of crashes at site i is a weighted average of estimated crash counts μ_i and actual crashes N_i .

B. Driver-Based Model

In this section, a Neural Network (NN) based driver-based model is introduced to predict risk probabilities using simulated data. Our selection of NN follows a study that we conducted comparing different models for this application. Traffic risks are associated with behaviors of the target vehicle and the interaction with other traffic streams. Besides, vehicles react differently in different traffic conditions and road sites such as intersections and segments. The proposed model considers those factors to account for the possible impact on traffic risks. Moreover, with crashes being rare events, we will use traffic conflicts as crash surrogates for developing a driver-based model. Traffic conflicts are influenced by a multitude of factors, such as roadway conditions, driver behavior, and vehicle condition. In actual applications, near-misses can be considered as traffic conflicts. In the study, we will predict conflict probabilities. The proposed NN model has three layers: the input layer, hidden layer, and output layer. There are fourteen nodes in the input layer and ten nodes in the hidden layer. The input set of the model includes five groups of variables: driver behaviors, roadway characteristics, vehicle statuses, and factors that are distance and speed-related. We will assume three types of driver behaviors, five roadway characteristics, vehicle queuing status, two distance-related measures, and three speed-related measures for model training. The introduced variables are able to explain exogenous

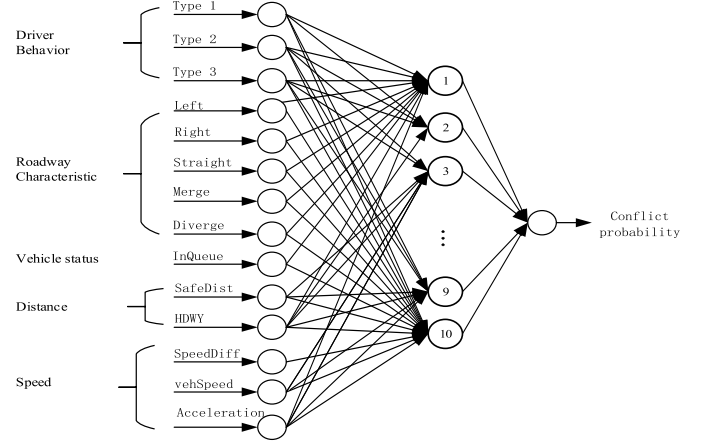


Fig. 2. The proposed Neural Network for Driver-based model.

influences such as speed difference, distance difference, and headway time between one vehicle and another other than a single vehicle. Roadway characteristics such as merge, diverge, and turning and queue status can also distinguish the impact from different road sites on risks. For instance, most turnings happen at intersections while moving straight often happens in segments. Besides, if a vehicle's in-queue status is true, it is safe to say the red traffic signal is on, and the vehicle is waiting at the intersection. A detailed description of the explanatory and response variables used in our study are listed in Appendix Table XI. The complete structure of the neural network is shown in Fig. 2.

The output of the model is the probability of conflict. We use the Sigmoid function, which outputs probability measures between 0 and 1 and takes input between $-\infty$ and $+\infty$. To update the weights and bias, gradient descent is used. In the output layer, the input, z , is given by:

$$z = \sum_{i=1}^{10} w_i x_i + b \quad (12)$$

where x_i is the output from the i th neuron in the hidden layer, w_i is the weight multiplied to the x_i , and b is the bias. The conflict probability, y , is given by the Sigmoid Function:

$$y = \frac{1}{1 + e^{-z}} \quad (13)$$

C. Integrated Risk Model

Finally, an Integrated Risk Model is introduced to compute a single risk measure from the estimated number of crashes and conflict probabilities. The model uses a fuzzy logic reasoning process. The use of fuzzy logic is motivated by its tolerance to imprecise data and the flexibility of handling complex data. It can model nonlinear relationships of arbitrary complexity by fuzzy rules. By user-defined fuzzy rules, the crash count and conflict probability obtained from the SPF model and driver-based model can be integrated into a single performance measure - risk score. Fig. 3 depicts the flow of the fuzzy logic reasoning process.

The input and output of the reasoning process are crisp numbers limited to a specific range. For instance, the range of the crash probability is $[0,1]$, and the range of the risk score is

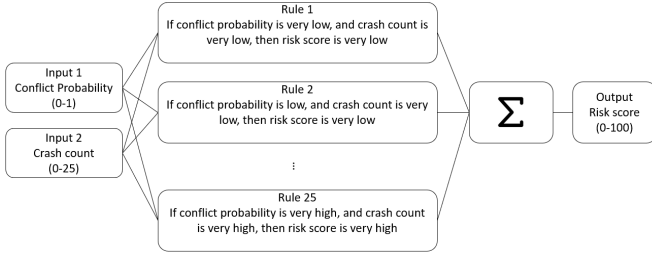


Fig. 3. The flow of the fuzzy logic reasoning process.

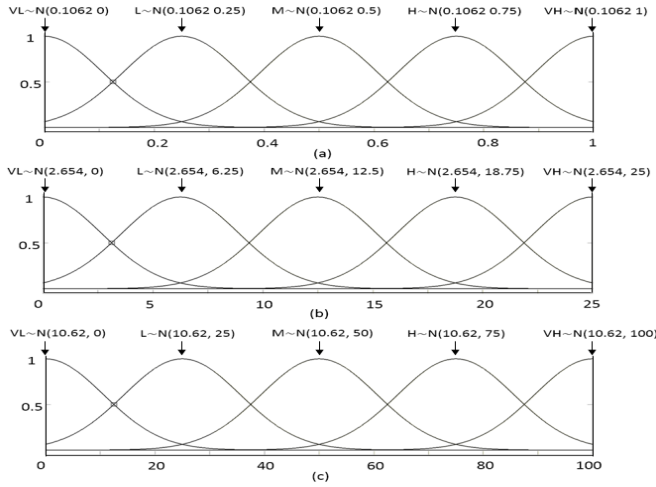


Fig. 4. Gaussian membership functions for: (a) conflict probability, (b) crash count, (c) risk score.

[0,100]. Then all the user-defined fuzzy rules are evaluated in parallel using fuzzy reasoning. Finally, the results of the rules are combined and defuzzified into a crisp output value. In the study, we use $\text{Min}(\cdot)$ operator for AND fuzzy operation and implication operation; $\text{Max}(\cdot)$ operator for OR fuzzy operation and aggregation operation. The Center of Area (COA) algorithm is used to find the center of the final aggregated shape for the crisp output value (defuzzification). Also, we denote $\tilde{CC}$ and $\tilde{CP}$ to be the fuzzy set of estimated crash counts and the probability of conflicts, respectively. The fuzzy output set, risk score, is denoted as $\tilde{RS}$. Gaussian membership functions are used for the fuzzification of input sets and defuzzification of the output set. Each fuzzy set has five membership functions that describe the probabilities of linguistic variables “very low” (VL), “low” (L), “medium” (M), “high” (H), and “very high” (VH), respectively. Fig. 4 (a) – (c) show the membership functions and corresponding standard deviation and mean for conflict probability, crash count, and risk score, respectively.

It is noted that any pair of input values (a, b) for conflict probability and crash count, yield fuzzy membership values $\text{VL}(a) \sim \text{VH}(a)$, $\text{VL}(b) \sim \text{VH}(b)$, respectively. Table II shows the matrix representation of all fuzzy rules for $\tilde{RS}$.

Row captions in the matrix contain the values of the conflict probability can take, column captions contain the values of the crash count can take, and each cell is the resulting risk score when the input variables take the values in that row and

TABLE II
THE FUZZY LOGIC REASONING RULE MATRIX

$\tilde{CC} \backslash \tilde{CP}$	VL	L	M	H	VH
VL	VL	VL	L	M	M
L	VL	L	L	M	M
M	L	L	M	M	H
H	M	M	M	H	VH
VH	M	M	H	VH	VH



Fig. 5. Historical crashes identified in road segments.

column. For instance, the cell (4, 2) can be read as follows: if the crash count is high, and the conflict probability is low, then the risk score is medium.

IV. ILLUSTRATIVE EXAMPLE

In this section, we demonstrate how crash counts are estimated by the developed SPF model using real crash data for sample roadway segments. We then use simulated data to train the NN model to predict conflict probabilities for the same roadway and show individual drivers’ risk profiles. Finally, we integrate the crash count estimates and conflict probabilities into risk scores and plot risk heat maps.

A. Prediction of Crash Count

Consider the rural roadway, as shown in Fig. 5. The roadway is divided into ten uniform segments. Each segment may have a unique set of physical attributes. As we will see later, one may need to further divide these segments due to the non-uniformity of risk scores. A total of 1,567 actual crashes were reported for this roadway between 2006 and 2017, with 840 rear-end crashes, 515 as crossing crashes, and 182 lane-change crashes.

In Fig. 5, dots represent crashes, and colors indicate the severity of crashes. A dot with a darker color means crashes with severe consequences. Note that a single dot may indicate more than one crash, depending on the geospatial resolution. Table III gives a comprehensive list of independent variables and sample values that can be used for training purposes.

Fig. 6 compares the actual and estimated crash counts by using the above SPF model from 2006 to 2017. Each time bucket includes all ten uniform rural roadway segments.

Using the Freeman-Tukey coefficient of determination ($R^2 = 0.99$), we see that the above prediction model is adequately able to explain the variation around crash counts.

TABLE III
VARIABLES FOR THE PROPOSED CRASH PREDICTION MODEL

Input	Variable name	Data type	Sample value
X_0	YEAR	INTEGER	2018
X_1	SEGMENT ID	INTEGER	8
X_2	SRI	INTEGER	18
X_3	FACILITY TYPE	CATEGORICAL	1
X_4	AREA TYPE	CATEGORICAL	2
X_5	SEGMENT LENGTH	DOUBLE	0.3
X_6	START-POINT	DOUBLE	0.00
X_7	END-POINT	DOUBLE	3.42
X_8	NUMBER OF LANES	INTEGER	3
X_9	ROAD TOTAL WIDTH	DOUBLE	40
X_{10}	SPEED LIMIT	INTEGER	60
X_{11}	AADT	DOUBLE	30,921
X_{12}	LANE WIDTH	DOUBLE	12
X_{13}	SHOULDER WIDTH	DOUBLE	8
X_{14}	SHOULDER TYPE	CATEGORICAL	1
X_{15}	PRESENCE OF MEDIAN	BINARY	0
X_{16}	MEDIAN WIDTH	DOUBLE	30
X_{17}	MEDIAN BARRIER	BINARY	0
X_{18}	PASSING LANE	INTEGER	0
X_{19}	2-WAY LEFT-TURN	BINARY	0
X_{20}	LIGHTING	BINARY	0
X_{21}	PRESENCE OF ON-STREET PARKING	BINARY	0
X_{22}	TYPE OF ON-STREET PARKING	BINARY	0
Output	Variable name	Data type	Sample value
Y	TOTAL NUMBER OF CRASHES	INTEGER	25

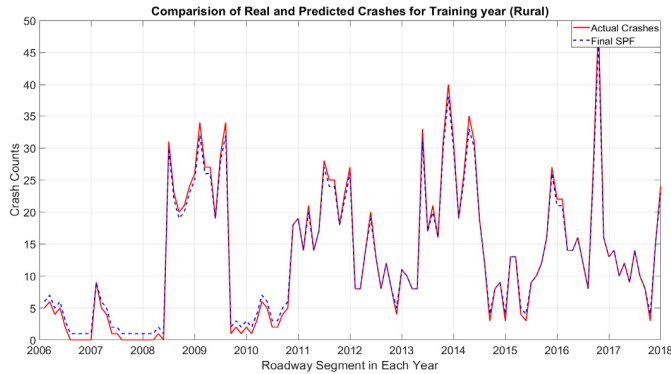


Fig. 6. The actual and estimated number of historical crashes.

TABLE IV
PREDICTED NUMBER OF CRASHES FOR 2018

Segment ID	1	2	3	4	5	6	7	8	9	10
Crashes counts	4	2	25	4	20	11	25	9	19	12

Table IV lists the predicted number of crashes in each of the segments for 2018 by the proposed SPF model.

The above analysis evaluates the risk of a roadway from a macro perspective by predicting the number of crashes. To assess the roadway from a micro perspective, the driver-based risk profile is created in the next section.

B. Risk Profile of Individual Drivers

In this section, we will demonstrate how driver-based conflict probability can be used to create risk profiles for

TABLE V
CONFLICT IDENTIFIED BY SSAM IN DIFFERENT SCENARIOS

id	Driver type	Volume	Rear-end	Crossing	Lane change
1	1	L1	186	5	7
2	1	L2	740	25	56
3	1	L3	5,048	58	301
4	1	Mixed	1861	30	130
5	2	L1	184	11	14
6	2	L2	745	20	53
7	2	L3	4,933	52	382
8	2	Mixed	1,857	32	133
9	3	L1	183	10	10
10	3	L2	764	19	66
11	3	L3	5,244	58	348
12	3	Mixed	1,853	33	130
13	mix	L1	172	7	16
14	mix	L2	1,103	20	54
15	mix	L3	4,456	70	307
16	mix	Mixed	1,857	28	160

TABLE VI
TRAINING VS. TESTING CONFUSION MATRIX

Training N=5526	Actual non-conflict	Actual conflict	
Predicted non-conflict	3846	533	87.8%
Predicted conflict	537	610	53.2%
	87.7%	53.4%	81.0%
Testing N=3684	Actual non-conflict	Actual conflict	
Predicted non-conflict	2554	377	87.1%
Predicted conflict	370	383	50.9%
	87.3%	50.4%	79.6%

individual drivers. We will then show how these profiles can be aggregated temporally and spatially. Traffic conflicts will be calculated using VISSIM and SSAM models of the roadway in Fig. 5. According to SSAM, there are three types of conflicts, namely Rear-end, Crossing, and Lane change. In addition, three different driver behavior models (i.e., Type 1, Type 2, and Type 3), used in a report by the Kentucky Transportation Cabinet [46], will be considered. Appendix Table IX and X define the car following driver behavior and lane change parameters used in the driver-based study, respectively. According to the two tables, observed vehicles, average standstill distance, additive and multiplicative parts of safety distance are the variables that define car-following behavior. Similarly, variables such as maximum deceleration and safety distance reduction factor distinguish lane changing behaviors. We assume three levels of traffic volume (L1: 400 vehicles/hour), (L2: 800 vehicles/hour) and (L3: 1,200 vehicles/hour). We consider a mixed case, which includes all driver types. Table V shows the conflicts by type from SSAM for a total of 16 simulation scenarios.

We divide a total of 9,210 selected observations from the simulation into a training dataset (60%) and a testing dataset (40%). Using the Neural Network of Fig. 2, we calculate the conflict probability concerning different driver types and traffic levels. Table VI compares training results and testing results.

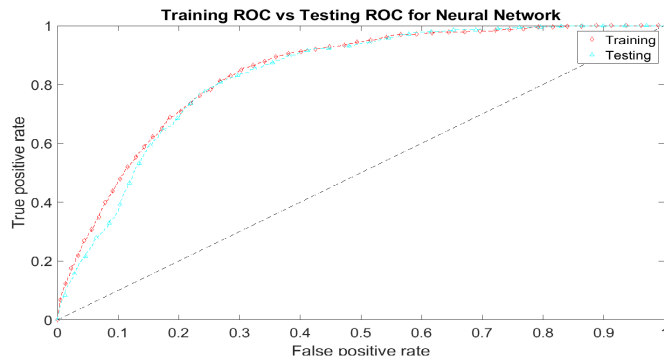


Fig. 7. Training ROC VS. Testing ROC.

The rows of the matrices are the predicted class, and the columns are the actual class. The diagonal cells represent correctly classified observations. The off-diagonal cells represent observations that are incorrectly classified. The last column of the plot shows the percentages of correct and incorrect predictions (positive predictive value and false discovery rate, respectively). The bottom row shows the percentages of classified and unclassified examples in each class. These metrics are often called the true positive rate and false-negative rate, respectively. We note that the overall prediction accuracy of training data is 81.0%, and testing data is 79.6%. Besides from the confusion matrix, the ROC curve also measures the goodness of the classification is. Fig. 7 shows the ROC curve of the Neural Network. AUC values for training and testing are 0.84 and 0.81, respectively.

An important remark is in order here. The above model does not have any inputs for weather conditions, vehicle conditions, and time of day. These could significantly influence traffic conflicts. The impact of these factors is difficult to demonstrate in a single simulation model, and as such, is omitted from the analysis. Thus, in this article, and only due to the use of simulation data, risks are not dependent on the time of day, nor are they dependent on vehicle types or weather conditions. In the absence of these factors, the traffic conditions and vehicle speed are the fast-changing inputs that could alter risks as one drives through a stretch of roadway. For a given driver type traveling through a roadway distance, a risk profile is then defined by a collection of conflict risks sampled at some constant time intervals (e.g., every one second).

To demonstrate the idea of risk profiles, a 700-foot long roadway subsection that includes an intersection is selected from Fig. 5. This sub-section is presented in Fig. 8.

For illustration purposes, we focus on the traffic from West to East. The subsection is further divided into three segments labeled as Link 1, Link 2, and Link 3. Running the trained Neural Network model for different traffic levels and driver behaviors, we compute the probability of conflicts at constant time intervals as a given driver passes through the roadway. Fig. 9 depicts the relationship between risk profiles and locations given a fixed time for Type 1 driver in L1 traffic.

Fig. 9 (a) gives a risk profile for a Type 1 driver passing through three differently colored links in L1 traffic. Fig. 9 (b) shows conflict risks for many Type 1 drivers who pass through



Fig. 8. Sample roadway segments.

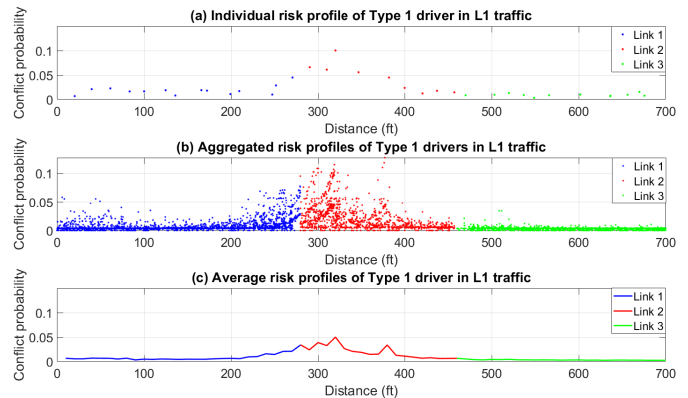


Fig. 9. Risk profile of the sample roadway section.

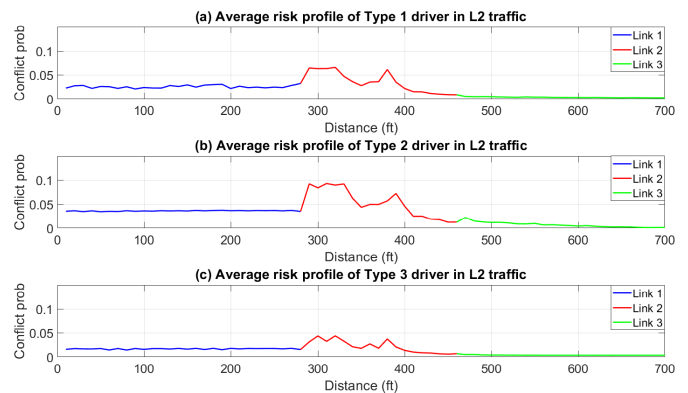


Fig. 10. Average risk profiles of different types of drivers in L2 traffic.

the three links. Finally, in Fig. 9 (c), we have average conflict risks for all sampled drivers. We note that these samples are taken within a given period on a given day. One can always repeat this sampling for the same period for many days to receive aggregate risk measures that span over drivers and days. Fig. 10 and 11 show how average risk profiles change with driver types and traffic conditions, respectively.

In all cases, Link 2 shows a higher conflict probability than others. As the traffic level changes, the overall risk profiles fluctuate. Besides, the Type 2 driver seems to be more conflict-prone. Next, we fuse crash count estimates with driver-based risk measures to obtain a more comprehensive risk heat map.



Fig. 11. Average risk profiles of Type 1 drivers in different traffic levels.

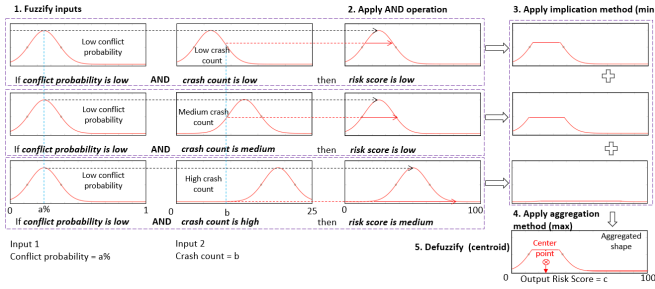


Fig. 12. Illustration of the fuzzy reasoning process.

C. Integration of Crash Data and Conflict Data

From the results of crash and conflict estimates, we set the range of membership functions for crash counts to $[0, 25]$ since the maximum number of predicted crashes in 2018 is 25 (Table IV), and the range of membership functions for conflict probability to $[0, 1]$. Inspired by safe driving Apps [47], [48], the range of membership functions for risk score is set to $[0, 100]$, with 100 being extremely risky. It is also a straightforward way to display the risk level for the driver. By defuzzifying the aggregated shape of pairwise comparisons for all the rules, risk scores are obtained. Fig. 12 illustrates the reasoning process of 3 sample fuzzy rules for an input pair (conflict probability = $a\%$, crash count = b).

We simulate and carry conflict analysis for the roadway in Fig. 5 for 5,400 simulation seconds. The simulation run is divided into 20 time slots, each with 270 seconds. The result is a 10×20 matrix where rows correspond to segment IDs and columns to the time slots. Consider Cell (10,3) for segment 3 and the time slot between 2,430 and 2,700 seconds. There is an estimate of 25 crashes (see Table IV) for this segment that stays the same for the simulation period since no traffic crash can be observed in our simulation model. For every driver who uses segment 3 during 2,430–2,700 seconds, we compute a conflict risk at constant time intervals. These samples constitute a risk profile of a driver stamped by segment ID. We recall that a risk profile is a collection of risk values sampled at constant time intervals as a driver goes a distance. So, in theory, we have many of these risk profiles that correspond to Cell (10,3). Fig. 13 gives risk scatter plots of Type 1 drivers for Cell (10,3) for L1 and L3 traffic volumes.

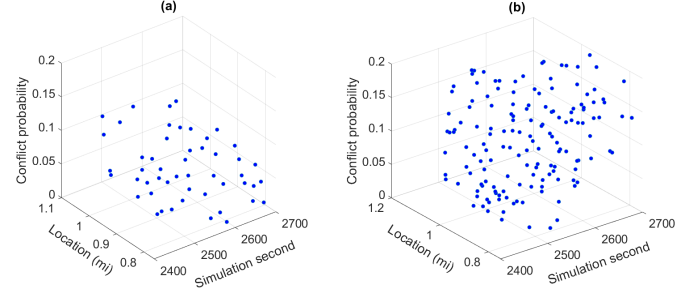


Fig. 13. Individual risk profiles of Type 1 driver in (a) L1 traffic and (b) L3 traffic.

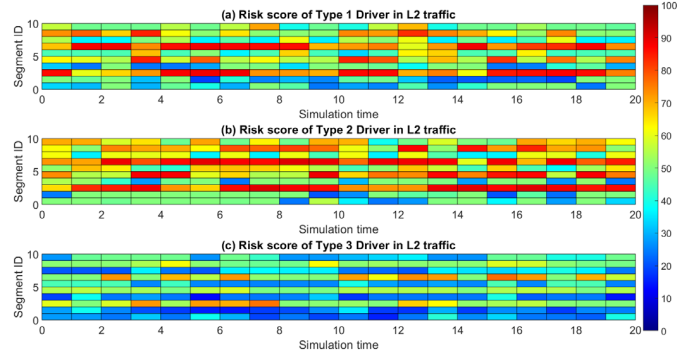


Fig. 14. Risk heat maps of different types of drivers in L2 traffic.

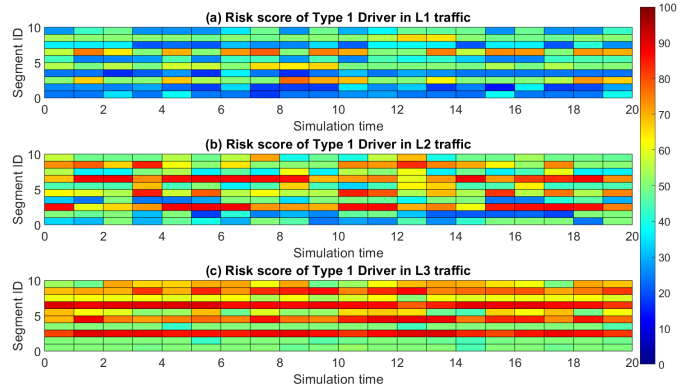


Fig. 15. Risk heat maps of Type 1 driver in different traffic levels.

According to these plots, the average and disparity of conflict risks for the same roadway segments and the corresponding periods increase with traffic volume.

Here, we will take a simple grand average of conflict risks in a given cell plot. Having driver-based average risk score and crash estimate for a cell, we can now run the fuzzy model. Fig. 14 and 15 show the risk score of different driver types in L2 traffic and Type 1 driver in different traffic levels, respectively.

The comparison of the same cells in Fig. 14 (a–c) yields interesting results with respect to the driver type. Type 2 drivers tend to have high risks since their scores are higher, while Type 3 drivers show low risks. The risk heat maps also reflect the impact of the crash count on different roadway segments. Associating risk heat maps with Table IV, we see

TABLE VII
CONFLICT THRESHOLDS USED IN OUR SIMULATION

Conflict-related parameters	Threshold value
Maximum time-to-collision (TTC)	1.5"
Maximum post-encroachment time	5"
Rear-end angle	30°
Crossing angle	80°

TABLE VIII
CRITERION OF DEFINING CONFLICT TYPE

Criterion	Conflict type
$0^\circ \leq \theta < \theta_1$	Rear-end
$\theta_1 \leq \theta < \theta_2$	Lane-change
$\theta_2 \leq \theta \leq 180^\circ$	Crossing

TABLE IX
CAR FOLLOWING DRIVER BEHAVIOR PARAMETERS

Car Following Parameters	Driver Behavior Model		
	Type 1	Type 2	Type 3
Look Ahead Distance (ft)	0 – 820.21	0 – 820.21	0 – 820.21
Observed Vehicles	6	4	4
Look Back Distance (ft)	0 – 492.13	0 – 492.13	0 – 492.13
Temporary Lack of Attention			
Duration (s)	1	1	1
Probability (%)	10	10	10
Average Standstill Distance (ft)	5.00	6.56	8.00
Additive Part of Safety Distance	1.50	2.60	2.50
Multiplicative Part of Safety Distance	2.50	3.60	3.50

high-risk scores also in high crash frequency segments such as 3, 5, 7, and 9. Besides, Fig. 15 depicts the correlation between traffic volumes and risk scores. L1 traffic results in the lowest risk score, and L3 traffic leads to the highest risk score.

V. CONCLUSION

In this paper, we developed a novel *Safe Route Mapping (SRM)* method, to score roadway safety using legacy crash based data and driver-based data that can be obtained from onboard devices and infrastructure sensors. An NB-based SPF model considering many roadway features is used to estimate the number of crashes, and the EB method is used to improve its performance. An NN model considering endogenous and exogenous factors from drivers and roadway is trained to predict vehicle conflicts dynamically. A fuzzy logic-based integration method is utilized to integrate different performance metrics into a single one. In the study, we first demonstrate how to develop the SPF model and use real crash data for crash prediction on sample roadway segments. Then, we show how to train the NN model with multiple sources of data such as roadway information, driver behaviors, and speed difference as inputs to predict conflict probabilities for the same roadway. Lastly, we illustrate the procedure to integrate crash estimates with conflict probabilities into a hybrid risk score. Due to

TABLE X
LANE CHANGE PARAMETERS

Lane Change Parameters	Driver Behavior Model		
	Type 1	Type 2	Type 3
Maximum Deceleration (ft/sec ²)			
Own	-16.00	-13.12	-10.00
Trailing Vehicle	-12.00	-9.84	-8.00
-1 ft/sec ² per distance (ft)			
Own	75	100	100
Trailing Vehicle	75	100	100
Accepted Deceleration (ft/sec ²)			
Own	-3.28	-3.28	-3.28
Trailing Vehicle	-3.28	-3.28	-3.28
Waiting Time Before Diffusion (sec)	60	60	60
Minimum Headway (front/ear) (ft)	1.64	1.64	1.64
Safety Distance Reduction Factor	0.55	0.70	0.65
Cooperative Lane Changed Allowed	No	No	No

TABLE XI
PREDICTOR AND RESPONSE VARIABLES USED FOR THE RISK MODEL

Variable Category	Independent Variable	Data Type	Description
Driver Behavior	Type1	bool	Type 1 driver
	Type2	bool	Type 2 driver
	Type3	bool	Type 3 driver
Roadway Characteristic	Left	bool	Left turn lane
	Right	bool	Right turn lane
	Straight	bool	Straight lane
	Merge	bool	Merge lane
	Diverge	bool	Diverge lane
Vehicle Status	InQueue	bool	Queue status
Distance	SafeDist	double	Safe distance
	HDWY	double	Headway distance
Speed	SpeedDiff	double	Speed difference between lead and host vehicles
	vehSpeed	double	Vehicle's actual speed
	Acceleration	double	Vehicle's acceleration
Variable category	Dependent Variable	Data type	Description
Conflict probability	Conflict	bool	Conflict probability

the lack of real-time driver-based data, we build a VISSIM simulation model to get vehicle trajectories for the case study and use the SSAM tool to obtain conflict data. We show that the SPF and the NN estimates are statistically sound. Sampling risk scores at constant time intervals as vehicles travel over a distance yield risk profiles, which are then used for risk heat mapping. Results show that road safety and drivers' risk profiles are associated with driver behaviors, road characteristics, and traffic conditions. The proposed *SRM* methodology is general and can be applied to real-life applications if actual data can be collected. However, the demonstration in this paper is only limited to driver-based data that can be obtained from simulation sources.

There are certain limitations that we can address in our future work. For instance, the developed SPF model can be applied to roadways in different states to test its transferability. The driver-based model can be extended to include more

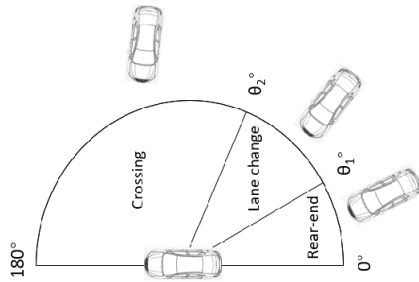


Fig. 16. Conflict angle in SSAM.

factors such as road facility types, road sites, and traffic signals for real-time traffic conflicts prediction. With additional data sources from onboard sensors equipped on future connected and automated vehicles (e.g., LiDAR, cameras, sensors), safe route recommendations for individual drivers can be realized using the risk heat maps generated by the *SRM* methodology.

APPENDIX

Tables VIII–XI and Fig. 16.

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REFERENCES

- [1] B. Coll, S. Moutari, and A. H. Marshall, "Hotspots identification and ranking for road safety improvement: An alternative approach," *Accident Anal. Prevention*, vol. 59, pp. 604–617, Oct. 2013.
- [2] M. A. Regan, J. D. Lee, and K. Young, *Driver Distraction: Theory, Effects, and Mitigation*. Boca Raton, FL, USA: CRC Press, 2008.
- [3] National Highway Traffic Safety Administration, "National motor vehicle crash causation survey: Report to congress," Nat. Highway Traffic Saf. Admin., Washington, DC, USA, Tech. Rep. DOT HS, 811, 059, 2008.
- [4] D. Ascone, T. T. Lindsey, and C. Varghese, "An examination of driver distraction as recorded in NHTSA databases," Nat. Highway Traffic Saf. Admin., Washington, DC, USA, Tech. Rep. DOT HS 811 216, 2009.
- [5] R. Taib, B. Itzstein, K. Yu, G. Mckellar, and G. Geers, "The future of driver assistance: Driver mental state monitoring," in *Proc. 21st ITS World Congr.*, Detroit, MI, USA, 2014.
- [6] Y. Dong, Z. Hu, K. Uchimura, and N. Murayama, "Driver inattention monitoring system for intelligent vehicles: A review," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 2, pp. 596–614, Jun. 2011.
- [7] Federal Motor Carrier Safety Administration, "Report to congress on the large truck crash causation study," U.S. Dept. Transp., Washington, DC, USA, Tech. Rep. MC-R/MC-RRA, 2006.
- [8] M. Warman. (May 9, 2013). *Road Buddy Mobile App Plots Safe Routes to Walk Home*. [Online]. Available: <https://www.telegraph.co.uk/technology/news/10047029/Road-Buddy-mobile-app-plots-safe-routes-to-walk-home.html>
- [9] S. T. Mineer, "Roadway safety analysis methodology," Ph.D. dissertation, Brigham Young Univ., Provo, UT, USA, 2016.
- [10] R. Srinivasan and D. Carter, "Development of safety performance functions for North Carolina," Res. Anal. Group, Dept. Transp., Boone, NC, USA, Tech. Rep. FHWA/NC/2010-09, 2011.
- [11] *H. S. Manual*, Amer. Assoc. State Highway Transp. officials (AASHTO), Washington, DC, USA, 2010, vol. 10.
- [12] V. V. Gayah and E. T. Donnell, "Establishing crash modification factors and their use," Mid-Atlantic Univ. Transp. Center, Tech. Rep. FHWA-PA-2014-005-PSU WO 6, 2014.
- [13] J. C. Comer, N. J. Rose, and L. S. Bombom, "Poisson regression analysis of highway fatality accident data in Oklahoma," *Int. J. Appl. Geospatial Res.*, vol. 5, no. 4, pp. 72–86, Oct. 2014.
- [14] H. Huang, H. Zhou, J. Wang, F. Chang, and M. Ma, "A multivariate spatial model of crash frequency by transportation modes for urban intersections," *Anal. Methods Accident Res.*, vol. 14, pp. 10–21, Jun. 2017.
- [15] W. Cheng, G. S. Gill, J. L. Ensich, J. Kwong, and X. Jia, "Multimodal crash frequency modeling: Multivariate space-time models with alternate spatiotemporal interactions," *Accident Anal. Prevention*, vol. 113, pp. 159–170, Apr. 2018.
- [16] F. Guo, X. Wang, and M. A. Abdel-Aty, "Modeling signalized intersection safety with corridor-level spatial correlations," *Accident Anal. Prevention*, vol. 42, no. 1, pp. 84–92, Jan. 2010.
- [17] B. Persaud, C. Lyon, and T. Nguyen, "Empirical Bayes procedure for ranking sites for safety investigation by potential for safety improvement," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1665, no. 1, pp. 7–12, Jan. 1999.
- [18] E. Hauer, "Overdispersion in modelling accidents on road sections and in empirical Bayes estimation," *Accident Anal. Prevention*, vol. 33, no. 6, pp. 799–808, Nov. 2001.
- [19] E. Hauer, D. W. Harwood, F. M. Council, and M. S. Griffith, "Estimating safety by the empirical Bayes method: A tutorial," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1784, no. 1, pp. 126–131, Jan. 2002.
- [20] R. Srinivasan and K. Bauer, "Safety performance function development guide: Developing jurisdiction-specific SPF's," FHWA, Washington, DC, USA, Tech. Rep. FHWA-SA-14-005, 2013.
- [21] A. Farid, M. Abdel-Aty, J. Lee, N. Eluru, and J.-H. Wang, "Exploring the transferability of safety performance functions," *Accident Anal. Prevention*, vol. 94, pp. 143–152, Sep. 2016.
- [22] T. A. Dingus *et al.*, "The 100-car naturalistic driving study: Phase II—Results of the 100-car field experiment," Nat. Highway Traffic Saf. Admin., Washington, DC, USA, Interim Project Rep. for DTNH22-00-C-07007, DOT HS 810 593, 2006.
- [23] S. G. Klauer, T. A. Dingus, V. L. Neale, J. D. Sudweeks, and D. J. Ramsey, "The impact of driver inattention on near-crash/crash risk: An analysis using the 100-car naturalistic driving study data," Nat. Highway Traffic Saf. Admin., Washington, DC, USA, Tech. Rep. DOT HS 810 594, 2006.
- [24] R. J. Hanowski, R. L. Olson, J. S. Hickman, and T. A. Dingus, "The 100-car naturalistic driving study: A descriptive analysis of light vehicle-heavy vehicle interactions from the light vehicle driver's perspective," Dept. Transp., Federal Motor Carrier Saf. Admin., Washington, DC, USA, Tech. Rep. FMCSA-RRR-06-004, 2006.
- [25] S. Herbel, L. Laing, and C. McGovern, "Highway safety improvement program (HSIP) manual," Federal Highway Admin., Washington, DC, USA, Tech. Rep. FHWA-SA-09-029, 2010.
- [26] V. Neale, T. Dingus, S. Klauer, J. Sudweeks, and M. Goodman, "An overview of the 100-car naturalistic driving study and findings. 2005," *ESV Paper*, Tech. Rep. 05-0400, 2005.
- [27] S. B. McLaughlin, J. M. Hankey, and T. A. Dingus, "A method for evaluating collision avoidance systems using naturalistic driving data," *Accident Anal. Prevention*, vol. 40, no. 1, pp. 8–16, Jan. 2008.
- [28] F. Guo and Y. Fang, "Individual driver risk assessment using naturalistic driving data," *Accident Anal. Prevention*, vol. 61, pp. 3–9, Dec. 2013.
- [29] K.-F. Wu and P. P. Jovanis, "Crashes and crash-surrogate events: Exploratory modeling with naturalistic driving data," *Accident Anal. Prevention*, vol. 45, pp. 507–516, Mar. 2012.
- [30] N. Arbabzadeh, M. Jafari, M. Jalayer, S. Jiang, and M. Kharbeche, "A hybrid approach for identifying factors affecting driver reaction time using naturalistic driving data," *Transp. Res. Part C, Emerg. Technol.*, vol. 100, pp. 107–124, Mar. 2019.
- [31] N. Arbabzadeh and M. Jafari, "A data-driven approach for driving safety risk prediction using driver behavior and roadway information data," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 2, pp. 446–460, Feb. 2018.
- [32] V. Carr and J. H. M. Tah, "A fuzzy approach to construction project risk assessment and analysis: Construction project risk management system," *Adv. Eng. Softw.*, vol. 32, nos. 10–11, pp. 847–857, Oct. 2001.
- [33] K. Wang, S. Zhao, J. N. Ivan, I. Ahmed, and E. Jackson, "Evaluation of hot spot identification methods for municipal roads," *J. Transp. Saf. Secur.*, vol. 12, no. 4, pp. 463–481, 2018.
- [34] T. Kim, "Assessment of vehicle-to-vehicle communication based applications in an urban network," Virginia Polytech. Inst. State Univ., Blacksburg, VA, USA, Tech. Rep., 2015.
- [35] Y. Huang, S. Jiang, M. Jafari, and P. Jin, "Infrastructure readiness for the anticipated transformative changes in transportation," in *Proc. Int. Conf. Smart Infrastruct. Construct. (ICSIC)*, Jan. 2019, pp. 481–489.

- [36] K. C. Dey, A. Rayamajhi, M. Chowdhury, P. Bhavsar, and J. Martin, "Vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication in a heterogeneous wireless network—performance evaluation," *Transp. Res. Part C, Emerg. Technol.*, vol. 68, pp. 168–184, Jul. 2016.
- [37] C.-W. You *et al.*, "CarSafe: A driver safety app that detects dangerous driving behavior using dual-cameras on smartphones," in *Proc. ACM Conf. Ubiquitous Comput.*, 2012, pp. 671–672.
- [38] R. Jabbar, K. Al-Khalifa, M. Kharbeche, W. Alhajyaseen, M. Jafari, and S. Jiang, "Real-time driver drowsiness detection for Android application using deep neural networks techniques," *Procedia Comput. Sci.*, vol. 130, pp. 400–407, 2018.
- [39] D. Gettman and L. Head, "Surrogate safety measures from traffic simulation models," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 1840, no. 1, pp. 104–115, Jan. 2003.
- [40] W. Young, A. Sobhani, M. G. Lenné, and M. Sarvi, "Simulation of safety: A review of the state of the art in road safety simulation modelling," *Accident Anal. Prevention*, vol. 66, pp. 89–103, May 2014.
- [41] C. Wang, M. A. Quddus, and S. G. Ison, "The effect of traffic and road characteristics on road safety: A review and future research direction," *Saf. Sci.*, vol. 57, pp. 264–275, Aug. 2013.
- [42] L. Pu, R. Joshi, and S. Energy, "Surrogate safety assessment model (SSAM)—software user manual," Turner-Fairbank Highway Res. Center, McLean, VA, USA, Tech. Rep. FHWA-HRT-08-050, 2008.
- [43] J. Xu, K. M. Kockelman, and Y. Wang, "Modeling crash and fatality counts along mainlanes and 1 frontage roads across Texas: 2 the roles of design, the built environment, and weather 3," in *Proc. 93rd Annu. Meeting Transp. Res.*, 2014, p. 24.
- [44] A. C. Cameron and P. K. Trivedi, *Regression Analysis of Count Data*. Cambridge, U.K.: Cambridge Univ. Press, 2013.
- [45] J. M. Hilbe, *Negative Binomial Regression*. Cambridge, U.K.: Cambridge Univ. Press, 2011.
- [46] C. Smith, "VISSIM development and calibration report: US 231 Scottsville Road scoping and traffic operations study," Kentucky Transp. Cabinet, Division Planning, Warren County, KY, USA, Tech. Rep. 3-8702.00, 2014.
- [47] D. Scapaticci, U. Fermi, and A. Saroldi, "Safety and comfort driver support functions in trip mate concept car," in *Proc. 7th World Congr. Intell. Syst.*, 2000.
- [48] B. Schwartz, *Driver Safety Scorecards: The First Step to a Safer Fleet*. Accessed: Oct. 8, 2020. [Online]. Available: <https://www.geotab.com/>



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